

CBSE Class 12 Physics — Half-Yearly Predicted Paper (Set 6 of 10)

Section A: 1-Mark Questions (MCQ/Assertion-Reason)

Q1. The electric potential V at a distance r from a point charge q is V . The magnitude of the electric field E at that same point is given by: (a) $E = V/r$ (b) $E = V \cdot r$ (c) $E = V/r^2$ (d) $E = V^2/r$ [PYQ 2021] | **[Chapter 2: Electrostatic Potential & Capacitance | Confidence: Very High]**

Q2. A parallel plate capacitor is charged by a battery and then the battery is disconnected. If the separation between the plates is now physically increased, the potential difference between the plates will: (a) Decrease (b) Increase (c) Remain constant (d) Become exactly zero [PYQ 2020] | **[Chapter 2: Electrostatic Potential & Capacitance | Confidence: High]**

Q3. The drift velocity of free electrons in a copper wire of length L is v_d . If the length of the wire is doubled ($2L$) while keeping the applied potential difference entirely constant, the new drift velocity will become: (a) v_d (b) $2v_d$ (c) $v_d/2$ (d) $v_d/4$ [Practice] | **[Chapter 3: Current Electricity | Confidence: Very High]**

Q4. An alpha particle and a proton enter a uniform transverse magnetic field with the exact same velocity. The ratio of the radii of their circular paths ($r_\alpha : r_p$) will be: (a) 1 : 1 (b) 1 : 2 (c) 2 : 1 (d) 4 : 1 [PYQ 2019] | **[Chapter 4: Moving Charges & Magnetism | Confidence: Very High]**

Q5. For an ideal diamagnetic substance, the magnetic susceptibility (χ) is: (a) Small and positive (b) Large and positive (c) Small and negative (d) Zero [PYQ 2020] | **[Chapter 5: Magnetism & Matter | Confidence: High]**

Q6. A solenoid has a self-inductance L . If the number of turns of the solenoid is doubled while keeping its physical length and area of cross-section constant, its new self-inductance will become: (a) $L/2$ (b) $2L$ (c) $4L$ (d) $L/4$ [PYQ 2017] | **[Chapter 6: Electromagnetic Induction | Confidence: Very High]**

Q7. In a series LCR alternating current circuit, the voltage measured across the inductor, capacitor, and resistor are 10 V, 10 V, and 20 V respectively. The total voltage of the AC source is: (a) 40 V (b) 20 V (c) 10 V (d) $20\sqrt{2}$ V [Practice] | **[Chapter 7: Alternating Current | Confidence: Very High]**

Q8. The ratio of the speed of X-rays to the speed of microwaves propagating in a vacuum is: (a) > 1 (b) < 1 (c) 1 (d) Depends on their respective frequencies [PYQ 2018] | **[Chapter 8: Electromagnetic Waves | Confidence: High]**

Q9. When a convex lens made of glass (refractive index 1.5) is completely immersed in water (refractive index 1.33), its focal length: (a) Increases (b) Decreases (c) Remains unchanged (d) Becomes negative [PYQ 2016] | **[Chapter 9: Ray Optics | Confidence: Very High]**

Q10. The phenomenon of interference of light is fundamentally based on the principle of: (a) Conservation of momentum (b) Conservation of energy (c) Superposition of waves (d) Quantum nature of light [Practice] | **[Chapter 10: Wave Optics | Confidence: High]**

Q11. Two independent light sources are considered strictly coherent if they continuously emit light waves having: (a) Same intensity (b) Same amplitude and same frequency (c) Same wavelength and a constant phase difference (d) Same speed in all media [PYQ 2015] | **[Chapter 10: Wave Optics | Confidence: Very High]**

Q12. Photons of energy 5 eV fall on the surface of a photosensitive metal whose work function is 3 eV . The required stopping potential to halt the fastest photoelectrons is: (a) 2 V (b) 8 V (c) 5 V (d) 3 V [PYQ 2019] | **[Chapter 11: Dual Nature of Radiation & Matter | Confidence: Very High]**

For Questions 13 to 16, two statements are given — one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer from the codes (a), (b), (c), and (d) as given below.

(a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is NOT the correct explanation of A. (c) A is true but R is false. (d) A is false and R is also false.

Q13. Assertion (A): Two electric lines of force can never cross each other. **Reason (R):** The electric field at any specific point in space has only one unique direction. [PYQ 2022] | **[Chapter 1: Electric Charges & Fields | Confidence: Very High]**

Q14. Assertion (A): The temperature coefficient of electrical resistance is strictly positive for pure metals. **Reason (R):** As the temperature of a metal increases, the relaxation time of free electrons colliding with lattice ions drastically decreases. [PYQ 2021] | **[Chapter 3: Current Electricity | Confidence: Very High]**

Q15. Assertion (A): An induced emf appears in a closed conducting loop only when the magnetic flux linked with it changes over time. **Reason (R):** Faraday's law of electromagnetic induction states that the magnitude of induced emf is directly proportional to the rate of change of magnetic flux. [Practice] | **[Chapter 6: Electromagnetic Induction | Confidence: High]**

Q16. Assertion (A): The focal length of a spherical concave mirror significantly changes when it is completely immersed in water. **Reason (R):** The focal length of a spherical mirror inherently depends heavily on the refractive index of the surrounding medium. [PYQ 2017] | **[Chapter 9: Ray Optics | Confidence: Very High]**

Section B: 2-Mark Questions (Very Short Answer)

Q17. An electric dipole of dipole moment $\vec{p}$ is placed completely inside a hollow closed spherical surface. Calculate the total net electric flux passing through this spherical surface. Justify your answer. [PYQ 2015] | **[Chapter 1: Electric Charges & Fields | Confidence: Very High]**

Q18. Define the term 'drift velocity' of charge carriers in a conductor. Write its SI unit. [PYQ 2018] | **[Chapter 3: Current Electricity | Confidence: High]**

Q19. A proton is moving horizontally along the positive x-axis in a region where a uniform magnetic field exists directed exactly along the positive y-axis. Determine the specific direction of the magnetic Lorentz force acting on the proton. [Practice] | **[Chapter 4: Moving Charges & Magnetism | Confidence: High]**

Q20. Define the term 'self-inductance' of a coil. Mention its SI unit. [PYQ 2019] | **[Chapter 6: Electromagnetic Induction | Confidence: Very High]**

Q21. What is the absolute rest mass of a photon? If the wavelength of a photon is λ , write the expression for its linear momentum. [PYQ 2016] | **[Chapter 11: Dual Nature of Radiation & Matter | Confidence: Very High]**

Section C: 3-Mark Questions (Short Answer)

Q22. Three point charges q_1 , q_2 , and q_3 are placed at the three vertices of an equilateral triangle of side a . Deduce the mathematical expression for the total electrostatic potential energy of this system of charges. [PYQ 2020] | **[Chapter 2: Electrostatic Potential & Capacitance | Confidence: Very High]**

Q23. State Kirchhoff's two rules for solving complex electrical circuits. Name the fundamental conservation principle on which each of these two rules is strictly based. [PYQ 2019] | **[Chapter 3: Current Electricity | Confidence: Very High]**

Q24. Calculate the exact magnitude of the magnetic field at the centre of a tightly wound circular coil of 100 turns, having a radius of 10 cm, and carrying a steady current of 5 A. (Take $\mu_0 = 4\pi \times 10^{-7} \text{ T m/A}$). [Practice] | **[Chapter 4: Moving Charges & Magnetism | Confidence: High]**

Q25. A jet plane is flying horizontally towards the West at a speed of 1800 km/h. What is the motional voltage difference developed between the ends of the wing having a span of 25 m, if the Earth's magnetic field at the location has a magnitude of $5 \times 10^{-4} \text{ T}$ and the local dip angle is 30° ? [PYQ 2018] | **[Chapter 6: Electromagnetic Induction | Confidence: Very High]**

Q26. State any two important physical properties of electromagnetic waves. Mention two distinct practical uses of Ultraviolet (UV) rays in daily life. [PYQ 2015] | **[Chapter 8: Electromagnetic Waves | Confidence: Very High]**

Q27. Define the 'critical angle' for a given pair of media. Derive the mathematical relation between the critical angle and the refractive index of the denser medium with respect to the rarer medium. [PYQ 2019] | **[Chapter 9: Ray Optics | Confidence: High]**

Q28. Monochromatic light of wavelength 500 nm falls on a photosensitive metal surface whose work function is 1.9 eV. Find the maximum kinetic energy of the emitted photoelectrons (in eV). (Take $hc = 1240 \text{ eV nm}$). [Practice] | **[Chapter 11: Dual Nature of Radiation & Matter | Confidence: Very High]**

Section D: 4-Mark Questions (Case-Study Based)

Q29. Case Study: Energy Stored in Capacitors Defibrillators are critical medical devices used to restore a normal heartbeat by sending a strong, brief electrical shock through the heart. They essentially use a large capacitor to store a massive amount of electrical energy, which can be discharged rapidly. Capacitors can be combined in electrical circuits in series or parallel to obtain a desired equivalent capacitance and energy storage capability. The equivalent capacitance of a series combination is smaller than the smallest individual capacitance, whereas in parallel, it is larger than the largest individual capacitance. (A) If two identical capacitors of capacitance C are connected strictly in series, what is their equivalent capacitance? (1 Mark) (B) When capacitors are

connected in series across a battery, which physical quantity remains completely constant across each of the capacitors? (1 Mark) (C) A $4\ \mu\text{F}$ capacitor and a $6\ \mu\text{F}$ capacitor are connected strictly in parallel across a $10\ \text{V}$ battery. Calculate the total equivalent capacitance and the total electrostatic energy stored in this specific combination. (2 Marks) [Practice] | **[Chapter 2: Electrostatic Potential & Capacitance | Confidence: Very High]**

Q30. Case Study: Young's Double Slit Experiment In 1801, Thomas Young brilliantly demonstrated the wave nature of light through his double-slit experiment. When monochromatic light from a single source illuminates two closely spaced narrow slits, the light waves emerging from these slits superimpose on a screen. This interference results in alternate bright and dark equally-spaced bands called fringes. For a stable pattern to be formed, the two sources of light must be strictly coherent. (A) Define the term 'coherent sources' in the context of wave optics. (1 Mark) (B) What is the exact path difference between the waves from the two slits at the position of the 3^{rd} dark fringe on the screen? (1 Mark) (C) If the physical separation between the two slits (d) is halved and the distance of the screen from the slits (D) is exactly doubled, what will happen to the linear fringe width (β)? Justify mathematically. (2 Marks) [PYQ 2022] | **[Chapter 10: Wave Optics | Confidence: Very High]**

Section E: 5-Mark Questions (Long Answer)

Q31. (A) Define 'Electric Dipole Moment'. Is it a scalar or a vector? (B) Derive the mathematical expression for the electric field intensity $\vec{E}$ at a point on the **axial line** of an electric dipole. (C) Derive the mathematical expression for the electric field intensity $\vec{E}$ at a point on the **equatorial line** of an electric dipole. (D) Show mathematically that for a very short dipole (where the distance r is much larger than the dipole length $2a$), the magnitude of the electric field on the axial line is exactly twice the magnitude of the electric field on the equatorial line at the same distance r . [PYQ 2018, 2020] | **[Chapter 1: Electric Charges & Fields | Confidence: Very High]**

Q32. (A) A series LCR circuit is connected to an alternating voltage source $V = V_0 \sin(\omega t)$. Using a phasor diagram, systematically derive the mathematical expression for the net impedance (Z) of the circuit and the phase angle (ϕ) by which the current leads or lags the voltage. (B) Define the term 'Electrical Resonance' in a series LCR circuit. (C) Obtain the expression for the resonant frequency (f_r) of the circuit. (D) What is the exact numerical value of the power factor of a series LCR circuit operating strictly at resonance? [PYQ 2019, 2023] | **[Chapter 7: Alternating Current | Confidence: Very High]**

Q33. (A) State Huygens' principle of secondary wavelets. (B) Draw a neatly labelled diagram showing the passage of a plane wavefront from an optically denser medium to an optically rarer medium. (C) Using this Huygens' geometrical construction, rigorously derive Snell's law of refraction ($\frac{\sin i}{\sin r} = \frac{v_1}{v_2} = \frac{\mu_2}{\mu_1}$). (D) In a Young's double-slit experiment setup, what will happen to the interference fringes on the screen if one of the two identical slits is completely covered with an opaque piece of dark paper? [PYQ 2015, 2020] | **[Chapter 10: Wave Optics | Confidence: Very High]**

Answer Key

Q1: (a) $E = V/r$. (For a point charge, $V = kq/r$ and $E = kq/r^2$. Thus, $E = V/r$). **Q2:** (b) Increase. (Battery is disconnected, so charge Q remains constant. Capacitance $C = \epsilon_0 A/d$. Increasing d decreases C . Since $V = Q/C$, a decrease in C results in an increase in V). **Q3:** (c) $v_d/2$. (Drift velocity $v_d = \frac{eE\tau}{m} = \frac{eV\tau}{mL}$. Thus, $v_d \propto 1/L$. Doubling length halves the drift velocity). **Q4:** (c) $2 : 1$. (Radius $r = mv/qB$. Ratio $r_\alpha/r_p = (m_\alpha/q_\alpha)/(m_p/q_p) = (4m_p/2e)/(m_p/e) = 2/1$). **Q5:** (c) Small and negative. (Diamagnetic substances develop a feeble magnetization opposite to the external field). **Q6:** (c) $4L$. (Self-inductance $L = \mu_0 n^2 Al = \mu_0 N^2 A/l$. Thus $L \propto N^2$. Doubling N makes it $4L$). **Q7:** (b) 20 V . ($V = \sqrt{V_R^2 + (V_L - V_C)^2} = \sqrt{20^2 + (10 - 10)^2} = \sqrt{20^2 + 0} = 20 \text{ V}$). **Q8:** (c) 1 . (All electromagnetic waves travel with the exact same speed $c = 3 \times 10^8 \text{ m/s}$ in a vacuum). **Q9:** (a) Increases. (The relative refractive index of glass drops when placed in water compared to air, which increases the focal length). **Q10:** (c) Superposition of waves. **Q11:** (c) Same wavelength and a constant phase difference. **Q12:** (a) 2 V . (From Einstein's eq: $K_{\max} = E - \Phi = 5 \text{ eV} - 3 \text{ eV} = 2 \text{ eV}$. Stopping potential $V_0 = K_{\max}/e = 2 \text{ V}$).

Q13: (a) Both A and R are true and R is the correct explanation of A. **Q14:** (a) Both A and R are true and R is the correct explanation of A. **Q15:** (a) Both A and R are true and R is the correct explanation of A. **Q16:** (d) A is false and R is also false. (Reflection from a mirror depends purely on geometry, not the refractive index of the medium, so its focal length $f = R/2$ remains perfectly unchanged).

Q17: Zero. An electric dipole consists of two equal and opposite charges ($+q$ and $-q$). Total enclosed charge $Q_{\text{enc}} = +q - q = 0$. By Gauss's Law, $\Phi = Q_{\text{enc}}/\epsilon_0 = 0$. **Q18:** The average uniform velocity with which free electrons get drifted towards the positive terminal of a conductor under the influence of an external electric field. SI Unit: m/s . **Q19:** Velocity $\vec{v}$ is along $+\hat{i}$. Magnetic field $\vec{B}$ is along $+\hat{j}$. Force $\vec{F} = q(\vec{v} \times \vec{B}) = e(\hat{i} \times \hat{j}) = e(\hat{k})$. Force is strictly directed along the positive z-axis. **Q20:** Self-inductance is the property of a coil by virtue of which it opposes any change in the strength of current flowing through it by inducing an opposing emf in itself. SI Unit: Henry (H). **Q21:** The rest mass of a photon is strictly zero. Its linear momentum is $p = h/\lambda$.

Q22: The total potential energy is the sum of the potential energies of all possible charge pairs: $U = \frac{1}{4\pi\epsilon_0} \left[\frac{q_1q_2}{a} + \frac{q_2q_3}{a} + \frac{q_3q_1}{a} \right]$. **Q23:** 1. Junction Rule ($\sum I = 0$): The algebraic sum of currents meeting at a junction is zero. Based strictly on the Law of Conservation of Charge. 2. Loop Rule ($\sum \Delta V = 0$): The algebraic sum of changes in potential around any closed loop is zero. Based strictly on the Law of Conservation of Energy. **Q24:**

$$B = \frac{\mu_0 NI}{2R} = \frac{(4\pi \times 10^{-7}) \times 100 \times 5}{2 \times 0.10} = \frac{2000\pi \times 10^{-7}}{0.20} = 10000\pi \times 10^{-7} = \pi \times 10^{-3} \text{ T} \approx 3.14 \times 10^{-3} \text{ T}$$

Q25: The vertical component of Earth's magnetic field $B_v = B \sin \delta = 5 \times 10^{-4} \sin(30^\circ) = 2.5 \times 10^{-4} \text{ T}$. Speed $v = 1800 \text{ km/h} = 1800 \times \frac{5}{18} = 500 \text{ m/s}$. Motional emf $\mathcal{E} = B_v lv = 2.5 \times 10^{-4} \times 25 \times 500 = 3.125 \text{ V}$. **Q26:** Properties: (1) They are transverse in nature. (2) They travel in a vacuum with the speed of light c . Uses of UV rays: (1) In water purifiers to kill bacteria. (2) In LASIK eye surgery. **Q27:** Critical angle is the angle of incidence in the denser medium for which the angle of refraction in the rarer medium is exactly 90° . By Snell's Law: $\mu_d \sin i_c = \mu_r \sin 90^\circ \implies \sin i_c = \mu_r/\mu_d$. If the rarer medium is air ($\mu_r = 1$), then

$\mu = 1/\sin i_c$. **Q28:** Energy of incident photon $E = hc/\lambda = 1240/500 = 2.48 \text{ eV}$. Maximum kinetic energy $K_{max} = E - W = 2.48 \text{ eV} - 1.9 \text{ eV} = 0.58 \text{ eV}$.

Q29: (A) Equivalent series capacitance $C_s = (C \times C)/(C + C) = C/2$. (B) The Electric Charge (Q) remains perfectly constant across all capacitors in series. (C) Parallel equivalent

$C_p = C_1 + C_2 = 4 + 6 = 10 \mu\text{F}$. Energy

$U = \frac{1}{2} C_p V^2 = \frac{1}{2} \times (10 \times 10^{-6}) \times (10)^2 = 5 \times 10^{-6} \times 100 = 5 \times 10^{-4} \text{ J}$. **Q30:** (A) Coherent sources are two sources of light that continually emit waves of identical frequency and maintain a strictly constant (or zero) phase difference over time. (B) Dark fringes occur at path difference $\Delta x = (2n - 1)\lambda/2$. For $n = 3$, $\Delta x = (6 - 1)\lambda/2 = 5\lambda/2$. (C) Fringe width $\beta = \lambda D/d$. If $d' = d/2$ and $D' = 2D$, new width $\beta' = \lambda(2D)/(d/2) = 4(\lambda D/d) = 4\beta$. The fringe width will exactly quadruple.

Q31: (A) Dipole moment is the product of the magnitude of either charge and the vector distance between them ($\vec{p} = q \times 2\vec{a}$). It is a vector quantity pointing from negative to positive charge. (B) Draw an axis with $-q$ and $+q$. Point P is at distance r from the centre.

$E_{axial} = E_{+q} - E_{-q} = \frac{kq}{(r-a)^2} - \frac{kq}{(r+a)^2} = \frac{kq(4ar)}{(r^2-a^2)^2} = \frac{k(2p)r}{(r^2-a^2)^2}$. (C) Draw an equatorial line at distance r . E_{+q} and E_{-q} resolve into vertical (which cancel) and horizontal components which add up. $E_{eq} = 2E \cos \theta = 2 \left(\frac{kq}{r^2+a^2} \right) \frac{a}{\sqrt{r^2+a^2}} = \frac{kp}{(r^2+a^2)^{3/2}}$. (D) For a short dipole ($r \gg a$), $E_{axial} \approx \frac{2kp}{r^3}$ and $E_{eq} \approx \frac{kp}{r^3}$. Therefore, $E_{axial} = 2E_{eq}$. **Q32:** (A) Draw phasor diagram with V_R

along current I , V_L leading by 90° , V_C lagging by 90° . Net voltage $V = \sqrt{V_R^2 + (V_L - V_C)^2}$.

Since $V = IZ$, $V_R = IR$, $V_L = IX_L$, $V_C = IX_C$, we get

$IZ = I\sqrt{R^2 + (X_L - X_C)^2} \Rightarrow Z = \sqrt{R^2 + (X_L - X_C)^2}$. Phase angle

$\tan \phi = (V_L - V_C)/V_R = (X_L - X_C)/R$. (B) Resonance is the condition when the current amplitude in the LCR circuit becomes completely maximum. This happens when $X_L = X_C$. (C) At resonance, $\omega L = 1/\omega C \Rightarrow \omega^2 = 1/LC \Rightarrow 2\pi f_r = 1/\sqrt{LC} \Rightarrow f_r = \frac{1}{2\pi\sqrt{LC}}$. (D) At resonance, $X_L = X_C$, so $Z = R$. Phase angle $\phi = 0^\circ$. Power factor $\cos \phi = \cos(0^\circ) = 1$. **Q33:**

(A) Every point on a given wavefront acts as a fresh source of new secondary spherical wavelets that spread out in all directions with the speed of light. The forward envelope of these wavelets yields the new wavefront. (B) Draw plane incident wavefront AB striking the denser-to-rarer interface. Draw secondary wavelet from A with radius $v_2 t$ (larger) and tangent CD representing the refracted wavefront. (C) In $\triangle ABC$, $\sin i = BC/AC = v_1 t/AC$. In $\triangle ADC$, $\sin r = AD/AC = v_2 t/AC$. Ratio $\frac{\sin i}{\sin r} = \frac{v_1 t/AC}{v_2 t/AC} = \frac{v_1}{v_2}$. Since $\mu = c/v \Rightarrow v \propto 1/\mu$, we get $\frac{\sin i}{\sin r} = \frac{\mu_2}{\mu_1}$, which is Snell's Law. (D) If one slit is covered by opaque paper, the superposition (interference) of waves from two slits ceases immediately. The interference fringes will completely disappear, leaving only a broad, uniformly lit single-slit diffraction pattern created by the single uncovered slit.